

CV

Personal Information:

Name: MOHAMMED ALAMEER

Nationality: Sudanese

Gender: Male

Degree: PhD Candidate (Mechanical Engineering)



Contact Information:

Address: Room 802, Unit 2, Building 14, Zhonghai ,No. 36 Jiangan Road, Qixing District, Guilin, Guangxi, China

Mobile No /WeChat No: [+86 195 0773 5910](tel:+8619507735910)

Email Address: alameermohammed@mails.guet.edu.cn

Google Scholar: <https://scholar.google.com/citations?user=BELhEgQAAAAJ&hl=en&oi=ao>

ResearchGate: <https://www.researchgate.net/profile/Mohammed-Alameer-10>

LinkedIn: <https://www.linkedin.com/in/mohammed-alameer/>

Research & Teaching Interests

- Dual mobile robotic systems, with a focus on rough-terrain and planetary exploration rovers,
- Reinforcement Learning-Based Control,
- Advanced Control, Robotics Simulation (MATLAB/Simulink, Simscape),
- Mechanical Design and Prototyping.

Academic Profile

PhD candidate in Mechanical Engineering and university lecturer with over seven years of teaching experience, supported by a strong background in industrial-scale mechanical systems. My research focuses on rough-terrain and planetary mobile robotics, rocker-bogie rover architectures, and reinforcement learning-based adaptive control. I have practical experience in high-fidelity MATLAB/Simulink and Simscape Multibody modeling, as well as experimental prototyping, testing, and validation. My academic profile combines mechanical design, intelligent control, simulation-based analysis, and research-led teaching. I am capable of delivering core mechanical engineering courses while contributing to robotics, mechatronics, and intelligent systems research. I am also actively developing SCI-indexed journal publications related to mobile robotics and adaptive control.

Education

PhD in Mechanical Engineering (Ongoing)

Guilin University of Electronic Technology (GUET), China

Research Focus: Dual Mode robotic systems, rocker-bogie rovers, climbing robot and RL-based adaptive PID control

MSc in Mechanical Engineering

Sudan University of Science and Technology, Sudan (2015–2018)

Master's Thesis: "A Proposed Solution for the Erosive Wear Problem in Garri-4 Power Plant Sand Conveying System"

BSc (Hons) in Mechanical Engineering

Bahri University, Sudan (2008–2014)

Academic & Teaching Experience

Lecturer Mechanical Engineering

Bahri University, College of Engineering, Sudan (2018–2022)

Courses Taught:

- Engineering Drawing and CAD Applications
- Mechanical Vibration Course
- Workshop Technology Course
- Thermodynamics I & II
- Heat Transfer
- HVAC Systems Design and Operation

Responsibilities:

- Delivered undergraduate courses in Thermal Engineering, Steam Turbines, HVAC Systems, Safety Engineering, and Mechanical Systems.
- Designed course syllabi, lecture materials, tutorials, and laboratory experiments.
- Supervised undergraduate graduation projects and mentored students in research methodology.
- Integrated industry case studies and simulation tools to enhance student learning outcomes.

Teaching Assistant Mechanical Engineering

Bahri University, Sudan (2014–2018)

Courses Assisted:

- Engineering Thermodynamics
- Fluid Mechanics
- Mechanics of Materials
- Manufacturing Processes
- Mechanical Design Fundamentals

Responsibilities:

- Assisted in teaching Thermodynamics, Fluid Mechanics, and Mechanical Design.
- Supported laboratory sessions, grading, tutorials, and student consultations.
- Participated in departmental academic and accreditation activities.

Research Experience

Robotics Design, Modelling, and Adaptive Control

PhD Research, Guilin University of Electronic Technology, China

- Designed and developed a six-wheel rocker-bogie rover platform inspired by planetary exploration rovers, with emphasis on rough-terrain mobility, passive suspension articulation, and obstacle-crossing capability.
- Developed a dual-mode robotic system integrating a ground rover with a detachable climbing mechanism for operation in limited-access environments such as trees, poles, and uneven outdoor terrain.
- Built high-fidelity simulation models using MATLAB/Simulink and Simscape Multibody to analyse rover dynamics, wheel-terrain interaction, suspension behaviour, and system stability under different terrain conditions.
- Designed and implemented reinforcement-learning-based adaptive PID control frameworks using DDPG for steering and velocity regulation of a six-wheel rocker-bogie rover.
- Conducted simulation-based validation and comparative analysis of conventional PID, baseline DDPG, and shaped-reward DDPG control strategies for trajectory tracking and motion stability.
- Supported experimental prototyping and validation of custom rover hardware, including mechanical assembly, sensor integration, encoder-based measurement, and terrain traversal testing.
- Prepared SCI-indexed journal manuscripts in mobile robotics, adaptive control, and mechanical system design, including work published in Journal of Mechanical Science and Technology and ongoing work under revision in robotics/control.

Industrial Experience

Senior Shift Operation Engineer

- Sudanese Thermal Power Generating Co. Ltd., Garri-4 Power Plant (2016–2022)
- Operation and supervision of CFB boilers, steam turbines, and auxiliary systems.
- Led critical grid synchronization during national blackout events.
- Implemented system-level improvements increasing operational efficiency.
- Trained and supervised technical staff, ensuring safety and reliability.

Technical Skills:

- Simulation & Control: MATLAB/Simulink, Simscape Multibody,
- Robotics & Dynamics: Mobile robots, rocker-bogie suspension, wheel-terrain interaction, kinematics & dynamics
- Design & CAD: AutoCAD, Revit (MEP), CATIA, SolidWorks
- Analysis & Modelling: COMSOL Multiphysics, multibody and contact modelling
- Mechanical Systems: Steam turbines, boilers (CFB), HVAC, MEP systems

Teaching, Research & Professional Training

Pedagogical & Academic Training

- Systems and Methods of University Teaching Khartoum University Center for Advanced Training (2018)
- Instructional Technology in University Teaching Khartoum University Center for Advanced Training (2018)
- Measurement and Evaluation in Higher Education Khartoum University Center for Advanced Training (2018)
- Basics of Scientific Research & Scientific Writing Khartoum University Center for Advanced Training (2018)
- Scientific Paper Writing Bahri University (2015)

Engineering & Professional Training

- Steam Turbine Troubleshooting Training Center for Integrated Development (2016)
- Boiler, Turbine & Auxiliaries: Procedures and Troubleshooting Garri-4 CFB Power Plant (2018)
- Primavera Project Management Software Training Center for Integrated Development (2021)
- Business Start-Up Program British Council (2015)
- Comprehensive Mechanical, Electrical, Plumbing Systems (MEP) (2022)

Publications:

- [1]Alameer, M., Yao, L., Liu, X. et al. ClimbNav-R: A dual-mode robot for seamless ground-to-vertical transitions in limited-access environments. J Mech Sci Technol (2026). <https://doi.org/10.1007/s12206-026-0451-2>.
- [2]ALAMEER M, AL-BARASHI M, JALLOW A, et al. Hybrid DDPG-based adaptive PID control for six-wheel rocker-bogie rover [J]. International Journal of Intelligent Robotics and Applications, 2026. <https://doi.org/10.1007/s41315-026-00568-1>.
- [3]Jallow A, Zhang F, Alameer M, Sun T, Chen Z, Liu X, Peng Y, Wangyang P, Zhu J, Shifeng X, Li H. Evaluating the mechanical stability and flexibility of Ga₂O₃ films on mica substrates via pulsed laser deposition and CASTEP simulations[J]. Journal of Crystal Growth, 2024, 648: 127900.
- [4]Al-Barashi M, Wang Y, Tang Z, Alameer M. Variable speed wind turbine control system with doubly-fed induction generator[C]. In: Proceedings of the 2025 International Conference on Electrical Automation and Artificial Intelligence (ICEAAI). IEEE, 2025: 361-367. <https://doi.org/10.1109/ICEAAI64185.2025.10956445>.
- [5]El-Hadi, Anwar, Ingi A. El-Hadi, and Mohammed AI Alameer. "The Normal: Minimising Energy Use; The Abnormal: Changing Habits." Mediterranean Green Buildings & Renewable Energy: Selected Papers from the World Renewable Energy Network's Med Green Forum. Cham: Springer International Publishing, 2016. https://doi.org/10.1007/978-3-319-30746-6_14.

Languages

- Arabic: Native
- English: Proficient (capable of teaching technical courses in English).
- Chinese: HSK4

References

Prof. Zupeng Zhou

Professor, School of Mechanical Engineering
Guilin University of Electronic Technology
Contact: +86 136 0773 7105, zupengzhou@163.com

Prof. Maged Al-Barashi

Professor, School of Aeronautics and Astronautics
Guilin University Aerospace Technology
Contact: +86 176 7703 1335, maged23064@guat.edu.cn

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